Conjecture 00000004105: incompatible growth clauses at an actual star

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1 The necessary conjunction

The original bilingual source defines a motion group using a graph's configuration space, then claims that its growth is polynomial if and only if the graph is a tree. The same conjecture describes the star-graph motion group as strictly super-polynomial (and sub-exponential). These two growth clauses cannot hold simultaneously, regardless of which candidate growth functions are assigned to the graphs.

We prove this universal obstruction, rather than assigning a value to the growth of a configuration-space fundamental group. The particle number and configuration-space conventions are unspecified in the source; the argument applies to any consistent choice giving a growth function for each relevant graph. No fundamental group or its actual growth is computed here. The sub-exponential and minimality assertions are unnecessary to the contradiction.

2 Actual graph and quantitative growth

Let S be the actual four-vertex star, with center 0 and vertex set $\{0, 1, 2, 3\}$. In Lean it is Mathlib's `SimpleGraph.starGraph (0 : Fin 4)`. The library's genuine `IsTree` theorem proves that S is a tree; its actual edge set has cardinality three.

For an arbitrary candidate $f : \mathbb{N} \to \mathbb{R}$, polynomial upper growth means

$$\exists d \in \mathbb{N}, \qquad f(n) = O(n^d) \quad (n \to \infty).$$

The code uses Mathlib's actual `IsBigO` and proves its equivalence to

$$\exists d,\ C > 0,\ N, \quad \forall n \geq N : \quad |f(n)| \leq Cn^d. \tag{1}$$

The quantitative super-polynomial condition is

$$\forall d \in \mathbb{N}\ \forall C > 0\ \forall N \in \mathbb{N}, \quad \exists n \geq N : \quad Cn^d < f(n). \tag{2}$$

It is defined through numerical values and thresholds, not as a syntactic negation. For non-negative count functions, the proof also shows that (2) is equivalent to the absence of every polynomial big-O upper bound. It does not demand the stronger condition of eventual domination at every subsequent index.

3 The contradiction and its scope

Suppose a candidate graph-to-growth assignment satisfies the source's tree-to-polynomial implication and the star's asserted (2). Since S is a tree, its candidate function f_S satisfies (1), with some actual d, C, N. Applying (2) to those same values gives $n \geq N$ with

$$Cn^d < f_S(n) \leq |f_S(n)| \leq Cn^d,$$

a contradiction. Both thresholds and constants belong to the same function; no numerical growth value is substituted.

The proof also covers every finite vertex type and every star center, so it does not depend on the star's unspecified order. The final Lean theorems quantify over every assignment to actual four-vertex graphs, then explicitly over assignments to all finite labelled graphs, including natural-valued candidate counting functions. Thus any actual motion-group growth assignment is covered if it is to satisfy both source clauses. This does not prove any standalone classification of graph configuration spaces, any triviality of tree motion groups, or any corrected conjecture omitting either clause.

4 Prior failed submission and verification

The contradiction idea was already public in PR 293. Its maintainer review rejected a Lean theorem consisting only of fixed natural-number arithmetic, without actual graph or growth objects. The present proof repairs those omissions using a real star/tree, arbitrary growth functions, actual big-O, and all numerical constants and cutoffs. We do not inherit the prior PR's unsupported claim that tree motion groups are trivial.

All named local declarations are audited in a pinned Lean 4.33.0 and Mathlib project. The package includes a new Main build, full official fresh kernel replay, source-bound command and log receipts, original bilingual source, LaTeX and this PDF. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted with a separate supervising AI reviewing the proof, source scope and rendered pages; no human certification or organizer acceptance is claimed.